DESHAN SIRIWARDHANA

Full-Stack Developer

BSc (Hons) Software Engineering | CINEC Campus

CONTACT

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EDUCATION

CINEC Campus

2024 - Present | Malabe, Sri Lanka
BSc (Hons) in Software Engineering
Key coursework: *Data Structures & Algorithms, Software Construction and Evolution, Software Quality Assurance, Database Management Systems, Mobile Computing, Advanced Web Technologies, Software Testing.*

Swarna Jayanthi College

2022 | Kegalle, Sri Lanka
G.C.E. Advanced Level - Physical Science Stream

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Java, Python, C#, PHP, C, Kotlin, Dart

Frontend & Mobile: Next.js, React.js, Flutter, Riverpod, Tailwind CSS, Bootstrap, HTML5, CSS3, React (Native)

Backend & APIs: Node.js, Express.js, RESTful APIs, WebRTC, ASP.NET Core,Socket.IO (WebSockets), Laravel, Google Maps APIs

Databases & Cloud: PostgreSQL, Supabase, MySQL, MSSQL, MongoDB, Firebase (Auth, Firestore, FCM)

Testing & QA: JUnit 5, Postman (API Testing), Apache JMeter, Selenium WebDriver, Requirements Traceability (RTM)

DevOps & Tools: Git, GitHub, Docker, Nginx, Vercel, VS Code, IntelliJ IDEA, Android Studio, Figma

Additional Exposure: Angular, Vue.js, Spring Boot, Oracle

OTHER SKILLS

Leadership & Management: Team Leadership, Agile / Scrum Practices, Project Planning, Cross-Functional Collaboration

Problem Solving: Analytical Thinking, Debugging, Critical Decision Making, Requirements Analysis

Communication & Interpersonal: Technical Documentation, Stakeholder Communication, Public Speaking, Client Engagement

Workplace Competencies: Time Management, Adaptability, Active Listening, Conflict Resolution

PROFILE

Versatile Full-Stack Software Engineer and Startup Co-Founder with a solid foundation bridging core web fundamentals (HTML5, CSS3, modern JavaScript) to advanced enterprise architectures (Next.js, React.js, Flutter, TypeScript, Node.js, .NET Core). Proven track record of turning requirements into production-ready web and mobile platforms integrating real-time WebSockets, AI/OCR services, and scalable cloud databases (PostgreSQL, Supabase, Firebase). Dedicated to building accessible, clean, and type-safe systems backed by automated QA testing and agile delivery.

EXPERIENCE

Co-Founder & Technical Lead

2026 - Present

CodeCraft Solutions

Co-founded a software solutions venture, directing technical strategy, system architecture, and modern UI/UX workflows for bespoke web and mobile applications.

Architected and shipped responsive full-stack solutions for local business clients, translating stakeholder requirements into scalable, maintainable software products.

FEATURED PROJECTS

ShelfLife AI - AI-Powered Inventory & Expiry Loss Prevention Platform

React.js (Vite) · Node.js · Firebase · Tesseract.js · REST API · PWA · JavaScript

- Built an AI-OCR scanning module using Tesseract.js to auto-extract expiration dates, cutting manual stock entry time by 90%.
- Engineered a real-time FEFO inventory engine with tiered alerts (30/7/2 days) and automated dynamic flash sales to eliminate expiry losses.
- Architected an offline-first PWA with background sync, role-based access, and an AI conversational assistant for inventory optimization.

SuraMaga - Zero-Hardware School Transport Safety & Fleet Management Platform

Group Project · In Progress · 2026

React Native ·· React.js · Tailwind CSS · Supabase (PostgreSQL) · Firebase Cloud Messaging · Google Maps API

- Contributing to a 4-member team building a mobile app and React admin portal for the All Island Inter District School Children Transport Association, using drivers' own smartphones instead of costly GPS and CCTV hardware.
- Specifying and designing core features: 3-stage QR "Silent Guard" anti-cheat checks, geofenced 500 m / 200 m parent alerts, 10 km SOS rescue handover and Google Maps smart route optimization.
- Designing an offline-first architecture (<50 KB auto-flushing trip cache, offline Guardian PIN verification) with Supabase Row Level Security that hides driver finances from admin accounts.

Kingswood Connect - Real-Time Student Attendance & Management Platform

React.js (Vite) · Node.js · Express.js · Firebase · Socket.IO · Tailwind CSS · JavaScript

- Engineered a hardware-free management platform enabling real-time smartphone QR scanning for desktop dashboards via Socket.IO WebSockets.
- Built a Node.js/Express backend integrated with Firebase (Firestore & Auth) to enforce multi-role authorization (Admin, Teacher, Student).
- Designed an administrative dashboard using React and Tailwind CSS for live attendance analytics, fee tracking, scheduling, and data recovery.

CERTIFICATIONS

Basics of Machine Learning Algorithms (Level 7)
Athena Global Education
Decision Trees, Naive Bayes, Predictive Modeling

Operate with DevOps
Microsoft
DevOps Practices, Continuous Integration & Delivery, Agile Workflows

Explore Microsoft 365 Security Foundations
Microsoft
Identity & Access Management, Threat Protection, Cloud Security

CIRS 2026 Workshop on Authentic Learning
CINEC Campus
Real-World Experiences That Build 21st-Century Skills

LANGUAGES

English: Professional Working Proficiency

Sinhala: Native / Bilingual Proficiency

LEADERSHIP

Active Member

AIESEC in CINEC | 2026 - Present

Contributing as an active member supporting youth leadership initiatives and chapter activities.

Student Member & Competitor

IEEE Student Branch of CINEC Campus
2026 - Present | Member ID: 102747783

Participating in **IEEEXtreme 20.0**, a 24-hour global competitive programming challenge, solving complex algorithmic and data structure problems.

Actively engaging in branch technical workshops, coding bootcamps, and professional networking sessions.

REFERENCES

Mr. D.P.N. Hasaranga
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Faculty of Computing, CINEC Campus
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Mr. Lahiru Nawarathne
Senior Tutor in Chemistry
Kingswood Education Centre
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AquaSoothe - Cross-Platform Wellness & Hydration Mobile App

Flutter · Dart · Riverpod · Firebase Auth · Node.js · Just Audio · REST APIs

- Smart Hydration Tracking: Tracks daily water intake, auto-adjusting targets based on real-time weather and medication schedules.
- AI Soundscapes: Generates custom relaxation and sleep tracks from text prompts using AI and on-device synthesis.
- Caregiver Monitoring: Enables remote monitoring of senior hydration and medication adherence with sleep correlation insights

KethPiyasa - Direct Farmer-to-Buyer B2B Agricultural Marketplace

Next.js · TypeScript · Node.js · Express.js · Firebase · Tailwind CSS

- Engineered a full-stack B2B marketplace featuring multi-role portals with real-time crop bidding, price negotiations, and map-based supplier discovery.
- Integrated an escrow-based secure payment workflow with automated invoice generation and QR code delivery verification.
- Architected a trilingual localization system (EN/SI/TA) paired with live economic center baseline price tracking for market transparency.

Neth-Sawan - AI-Powered Assistive Web Platform

React.js (Vite) · JavaScript · Node.js · WebRTC · Google Generative AI SDK · Firebase

- Built a real-time assistive communication portal for hearing-impaired users using WebRTC and Firebase signaling servers for peer-to-peer video streaming.
- Integrated Google Generative AI SDK to convert live camera feeds into instant visual contextual descriptions and scannable ambient alerts.
- Designed an accessible, high-contrast user interface compliant with WCAG accessibility guidelines to ensure seamless usability.

RIGA Academy - Dance Academy Management & Online Enrollment Platform

Next.js · TypeScript · Firebase (Auth & Firestore) · Tailwind CSS · Framer Motion

- Built a full-stack academy portal featuring an online enrollment workflow with package selection, slip uploads, and live status tracking.
- Configured Firebase Auth and Firestore for multi-role security (Admin/Student) and developed an admin dashboard for attendance, scheduling, and payment approvals.
- Crafted a modern, responsive UI/UX using Framer Motion and Tailwind CSS featuring interactive class schedules, media galleries, and dynamic animations.